

Metric for Performance Evaluation :

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★ Confusion Matrix

Actual \ Predicted	0	1
0	TN	FP
1	FN	TP

$$\rightarrow \text{Accuracy} = \frac{TP + TN}{P + N} \quad \text{Error} = \frac{FN + FP}{P + N} = 1 - \text{Accuracy}$$

$$\rightarrow \text{Sensitivity} = \frac{TP}{P} \quad \text{Specificity} = \frac{TN}{N}$$

(the prop of pos tuples that are correctly identified) (the prop of neg tuples that are correctly identified)

$$\rightarrow \text{Precision} = \frac{TP}{TP + FP} \quad \text{Recall} = \frac{TP}{TP + FN}$$

Ratio of correctly predicted pos / total no of tup predicted as positive

Ratio of true pos / total no of actual positive instance

★ F1 Score ($0 \leq F1 \leq 1$)

0 = Worst

1 = Best

F1 score gives harmonic mean of Precision and Recall (balanced measure of FP and FN)

★ F β Score

$\beta = 1 \Rightarrow$ Precision and Recall equally.

$\beta > 1 \Rightarrow$ Recall gets more importance

$0 < \beta < 1 \Rightarrow$ Precision gets more imp

Weighted measure of precision and recall

★ Note

- Why Harmonic mean : It's used to calculate mean of ratios and heavily punishes the lower value